2.54 Polar Differentiation

Worked Solutions

1 a)

$$r = 4\cos\theta \Rightarrow r\sin\theta = 4\cos\theta\sin\theta \quad \text{so } y = 4\cos\theta\sin\theta$$
$$r = 4\cos\theta \Rightarrow r\cos\theta = 4\cos^2\theta \quad \text{so } x = 4\cos^2\theta$$
$$\therefore y = 4\cos\theta\sin\theta \text{ and } x = 4\cos^2\theta$$

1 b)

$$r = 8\sin\theta \Rightarrow r\sin\theta = 8\sin^2\theta \quad \text{so } y = 8\sin^2\theta$$
$$r = 8\sin\theta \Rightarrow r\cos\theta = 8\sin\theta\cos\theta \quad \text{so } x = 8\sin\theta\cos\theta$$
$$\therefore y = 8\sin^2\theta \text{ and } x = 8\sin\theta\cos\theta$$

1 c)

$$r = 3 + 3\cos\theta \Rightarrow r\sin\theta = 3\sin\theta + 3\sin\theta\cos\theta \quad \text{so } y = 3\sin\theta + 3\sin\theta\cos\theta$$
$$r = 3 + 3\cos\theta \Rightarrow r\cos\theta = 3\cos\theta + 3\cos^2\theta \quad \text{so } x = 3\cos\theta + 3\cos^2\theta$$
$$\therefore y = 3\sin\theta + 3\sin\theta\cos\theta \text{ and } x = 3\cos\theta + 3\cos^2\theta$$

1 d)

$$r = 6 - 6\sin\theta \Rightarrow r\sin\theta = 6\sin\theta - 6\sin^2\theta \quad \text{so } y = 6\sin\theta - 6\sin^2\theta$$
$$r = 6 - 6\sin\theta \Rightarrow r\cos\theta = 6\cos\theta - 6\sin\theta\cos\theta \quad \text{so } x = 6\cos\theta - 6\sin\theta\cos\theta$$
$$\therefore y = 6\sin\theta - 6\sin^2\theta \text{ and } x = 6\cos\theta - 6\sin\theta\cos\theta$$

1 e)

$$r = 5\sin 2\theta \Rightarrow r\sin\theta = 5\sin 2\theta\sin\theta \quad \text{so } y = 10\cos\theta\sin^2\theta$$
$$r = 5\sin 2\theta \Rightarrow r\cos\theta = 5\sin 2\theta\cos\theta \quad \text{so } x = 10\sin\theta\cos^2\theta$$
$$\therefore y = 10\cos\theta\sin^2\theta \text{ and } x = 10\sin\theta\cos^2\theta$$

2 a)

$$\frac{dy}{d\theta} = -3\sin\theta - 6\cos\theta\sin\theta$$

2 b)

$$\frac{dy}{d\theta} = 5\cos\theta - 10\sin\theta\cos\theta$$

2 c)

$$\frac{dy}{d\theta} = -7\sin\theta + 14\cos\theta\sin\theta$$

2 d)

$$\frac{dy}{d\theta} = -2\cos\theta - 4\sin\theta\cos\theta$$

2 e)

$$\frac{dy}{d\theta} = -24\cos^2\theta\sin\theta + 4\sin\theta$$

2 f)

$$\frac{dy}{d\theta} = -6\sin^3\theta + 12\sin\theta\cos^2\theta$$

3 a)

$$r = 2(1+\sin\theta) \Rightarrow r\sin\theta = 2\sin\theta + 2\sin^2\theta$$
$$y = 2\sin\theta + 2\sin^2\theta$$
$$\frac{dy}{d\theta} = 2\cos\theta + 4\sin\theta\cos\theta = 2\cos\theta + 2\sin 2\theta$$

3 b)

$$r = 2(1+\sin\theta) \Rightarrow r\cos\theta = 2\cos\theta + 2\sin\theta\cos\theta$$
$$x = 2\cos\theta + \sin 2\theta$$
$$\frac{dx}{d\theta} = -2\sin\theta + 2\cos 2\theta$$

3 c)

$$2\cos\theta + 2\sin 2\theta = 2\cos\theta + 4\sin\theta\cos\theta = 2\cos\theta(1+2\sin\theta) = 0$$
$$\text{So } \cos\theta = 0 \text{ or } \sin\theta = -\frac{1}{2}$$
$$\therefore \theta = -\frac{\pi}{2}, \quad \theta = \frac{\pi}{2}, \quad \theta = -\frac{\pi}{6} \text{ or } \theta = -\frac{5\pi}{6}$$

3 d)

$$-2\sin\theta + 2\cos 2\theta = -2\sin\theta + 2(1 - 2\sin^2\theta) = 0$$
$$2\sin^2\theta + \sin\theta - 1 = (2\sin\theta - 1)(\sin\theta + 1) = 0$$
$$\text{So } \sin\theta = \frac{1}{2} \text{ or } \sin\theta = -1$$
$$\therefore \theta = \frac{\pi}{6}, \quad \theta = \frac{5\pi}{6} \text{ or } \theta = -\frac{\pi}{2}$$

3 e)

When $\theta = -\frac{\pi}{2}$, both $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} = 0$, so this is the value of θ for which there is a cusp point.

4 a)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = 2 + 2\cos\theta \Rightarrow y = r\sin\theta = 2\sin\theta + 2\sin\theta\cos\theta$$
$$\frac{dy}{d\theta} = 2\cos\theta - 2\sin^2\theta + 2\cos 2\theta$$
$$r = 2 + 2\cos\theta \Rightarrow x = r\cos\theta = 2\cos\theta + 2\cos^2\theta$$
$$\frac{dx}{d\theta} = -2\sin\theta - 4\cos\theta\sin\theta$$

Consider $\dfrac{dy}{d\theta} = 0$:

$$2\cos\theta - 2(1 - \cos 2\theta) + 2\cos 2\theta = 0$$
$$4\cos^2\theta + 2\cos\theta - 2 = 0$$
$$(\cos\theta + 1)(2\cos\theta - 1) = 0$$
$$\therefore \theta = \pi, \quad \theta = \frac{\pi}{3} \text{ or } \theta = -\frac{\pi}{3}$$

When $\theta = \pi$, $y = 0$ and $\dfrac{dx}{d\theta} = 0$, so this is a cusp. At $\theta = \frac{\pi}{3}$ and $-\frac{\pi}{3}$, respectively, $y = \frac{3\sqrt{3}}{2}$ and $y = -\frac{3\sqrt{3}}{2}$. So the tangent lines are $y = \frac{3\sqrt{3}}{2}$ and $y = -\frac{3\sqrt{3}}{2}$.

4 b)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = 3 - 3\sin\theta \Rightarrow y = r\sin\theta = 3\sin\theta - 3\sin^2\theta$$
$$\frac{dy}{d\theta} = 3\cos\theta - 6\cos\theta\sin\theta$$
$$r = 3 - 3\sin\theta \Rightarrow x = r\cos\theta = 3\cos\theta - 3\sin\theta\cos\theta$$
$$\frac{dx}{d\theta} = -3\sin\theta + 3\sin^2\theta - 3\cos 2\theta$$

Consider $\dfrac{dy}{d\theta} = 0$: $3\cos\theta(1 - 2\sin\theta) = 0$. Thus $\cos\theta = 0$ or $\sin\theta = \frac{1}{2}$, giving $\theta = \frac{\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$. At $\theta = \frac{\pi}{2}$, $y = 0$ but $\dfrac{dx}{d\theta} = 0$, so this is a cusp. At $\theta = -\frac{\pi}{2}$, $y = -6$. At $\theta = \frac{\pi}{6}$ and $\frac{5\pi}{6}$, $y = \frac{3}{4}$. So the tangent lines are $y = -6$ and $y = \frac{3}{4}$.

4 c)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = 4\sin 2\theta \Rightarrow y = r\sin\theta = 4\sin\theta\sin 2\theta$$
$$\frac{dy}{d\theta} = 8\sin\theta\cos 2\theta + 4\cos\theta\sin 2\theta$$
$$r = 4\sin 2\theta \Rightarrow x = r\cos\theta = 4\sin 2\theta\cos\theta$$
$$\frac{dx}{d\theta} = -4\sin 2\theta\sin\theta + 8\cos 2\theta\cos\theta$$

Setting $\dfrac{dy}{d\theta} = 0$ gives $8\sin\theta(2\cos^2\theta - \sin^2\theta) = 0$. Thus $\sin\theta = 0$ or $\tan^2\theta = 2$. At $\theta = 0, \pi$, $y = 0$ and $\dfrac{dx}{d\theta} \neq 0$. For $\tan\theta = \sqrt{2}$, $\sin\theta = \sqrt{2/3}$ and $\cos\theta = 1/\sqrt{3}$, giving $y = \frac{16\sqrt{3}}{9}$. For $\tan\theta = -\sqrt{2}$, $y = -\frac{16\sqrt{3}}{9}$. So the tangent lines are $y = 0$, $y = \frac{16\sqrt{3}}{9}$ and $y = -\frac{16\sqrt{3}}{9}$.

4 d)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = a + a\sin\theta \Rightarrow y = a\sin\theta + a\sin^2\theta$$
$$\frac{dy}{d\theta} = a\cos\theta + 2a\sin\theta\cos\theta$$
$$r = a + a\sin\theta \Rightarrow x = a\cos\theta + \frac{a}{2}\sin 2\theta$$
$$\frac{dx}{d\theta} = -a\sin\theta + a\cos 2\theta$$

Solving $\dfrac{dy}{d\theta} = 0$ gives $\cos\theta(1 + 2\sin\theta) = 0$, so $\cos\theta = 0$ or $\sin\theta = -\frac{1}{2}$. The values are $\theta = \frac{\pi}{2}, -\frac{\pi}{2}, -\frac{\pi}{6}, -\frac{5\pi}{6}$. At $\theta = -\frac{\pi}{2}$, $y = 0$ and $\dfrac{dx}{d\theta} = 0$, so there is a cusp. At $\theta = \frac{\pi}{2}$, $y = 2a$, and at the two values with $\sin\theta = -\frac{1}{2}$, $y = -\frac{a}{4}$. Therefore the tangent lines are $y = 2a$ and $y = -\frac{a}{4}$.

4 e)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = a - a\cos\theta \Rightarrow y = a\sin\theta - \frac{a}{2}\sin 2\theta$$
$$\frac{dy}{d\theta} = a\cos\theta - a\cos 2\theta$$
$$r = a - a\cos\theta \Rightarrow x = a\cos\theta - a\cos^2\theta$$
$$\frac{dx}{d\theta} = -a\sin\theta + 2a\cos\theta\sin\theta$$

Setting $\dfrac{dy}{d\theta} = 0$ gives $a(\cos\theta - 1)(2\cos\theta + 1) = 0$, so $\cos\theta = 1$ or $\cos\theta = -\frac{1}{2}$. Thus $\theta = 0, -\frac{2\pi}{3}, \frac{2\pi}{3}$. When $\theta = 0$, $y = 0$ but $\dfrac{dx}{d\theta} = 0$, so there is a cusp. At $\theta = -\frac{2\pi}{3}$ and $\frac{2\pi}{3}$, $y = -\frac{3\sqrt{3}}{4}a$ and $y = \frac{3\sqrt{3}}{4}a$. So the tangent lines are $y = -\frac{3\sqrt{3}}{4}a$ and $y = \frac{3\sqrt{3}}{4}a$.

4 f)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = a\cos 2\theta \Rightarrow y = a\sin\theta\cos 2\theta = a\sin\theta - 2a\sin^3\theta$$
$$\frac{dy}{d\theta} = a\cos\theta - 6a\sin^2\theta\cos\theta$$
$$r = a\cos 2\theta \Rightarrow x = 2a\cos^3\theta - a\cos\theta$$
$$\frac{dx}{d\theta} = -6a\cos^2\theta\sin\theta + a\sin\theta$$

Setting $\dfrac{dy}{d\theta} = 0$ gives $a\cos\theta(1 - 6\sin^2\theta) = 0$. Hence $\cos\theta = 0$ or $\sin\theta = \pm\frac{1}{\sqrt{6}}$. At $\theta = -\frac{\pi}{2}$ and $\frac{\pi}{2}$, $y = a$ and $y = -a$. For $\sin\theta = \frac{1}{\sqrt{6}}$, $y = \frac{\sqrt{6}}{9}a$; for $\sin\theta = -\frac{1}{\sqrt{6}}$, $y = -\frac{\sqrt{6}}{9}a$. So the tangent lines are $y = a$, $y = -a$, $y = \frac{\sqrt{6}}{9}a$ and $y = -\frac{\sqrt{6}}{9}a$.

4 g)

A tangent parallel to the initial line $\theta = 0$ requires $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = e^\theta \Rightarrow y = e^\theta\sin\theta, \quad \frac{dy}{d\theta} = e^\theta\cos\theta + e^\theta\sin\theta$$
$$r = e^\theta \Rightarrow x = e^\theta\cos\theta, \quad \frac{dx}{d\theta} = -e^\theta\sin\theta + e^\theta\cos\theta$$
$$e^\theta(\cos\theta + \sin\theta) = 0 \Rightarrow \tan\theta = -1 \Rightarrow \theta = -\frac{\pi}{4} \text{ or } \theta = \frac{3\pi}{4}$$
$$y\left(-\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}e^{-\pi/4}, \qquad y\left(\frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2}e^{3\pi/4}$$

So the tangent lines are $y = -\frac{\sqrt{2}}{2}e^{-\pi/4}$ and $y = \frac{\sqrt{2}}{2}e^{3\pi/4}$.

5 a)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = 4 - 4\cos\theta \Rightarrow x = 4\cos\theta - 4\cos^2\theta, \quad \frac{dx}{d\theta} = -4\sin\theta + 8\cos\theta\sin\theta$$
$$r = 4 - 4\cos\theta \Rightarrow y = 4\sin\theta - 2\sin 2\theta, \quad \frac{dy}{d\theta} = 4\cos\theta - 4\cos 2\theta$$
$$4\sin\theta(2\cos\theta - 1) = 0 \Rightarrow \theta = 0, \pi, -\frac{\pi}{3}, \frac{\pi}{3}$$

When $\theta = 0$, $x = 0$ but $\dfrac{dy}{d\theta} = 0$, so this is a cusp. At $\theta = \pi$, $x = -8$; at $\theta = \pm\frac{\pi}{3}$, $x = 1$. So the tangent lines are $x = -8$ and $x = 1$.

5 b)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = 2 + 2\sin\theta \Rightarrow x = 2\cos\theta + \sin 2\theta, \quad \frac{dx}{d\theta} = -2\sin\theta + 2\cos 2\theta$$
$$r = 2 + 2\sin\theta \Rightarrow y = 2\sin\theta + 2\sin^2\theta, \quad \frac{dy}{d\theta} = 2\cos\theta + 4\sin\theta\cos\theta$$

$$-2\sin\theta + 2\cos 2\theta = -2(2\sin\theta - 1)(\sin\theta + 1) = 0$$
$$\theta = \frac{\pi}{6}, \quad \theta = \frac{5\pi}{6} \quad \text{or} \quad \theta = -\frac{\pi}{2}$$

At $\theta = \frac{\pi}{6}$ and $\frac{5\pi}{6}$, $x = \frac{3\sqrt{3}}{2}$ and $x = -\frac{3\sqrt{3}}{2}$. At $\theta = -\frac{\pi}{2}$, $x = 0$ but $\dfrac{dy}{d\theta} = 0$, so this is a cusp. So the tangent lines are $x = \frac{3\sqrt{3}}{2}$ and $x = -\frac{3\sqrt{3}}{2}$.

5 c)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = 2\cos 2\theta \Rightarrow x = 2\cos\theta\cos 2\theta, \quad \frac{dx}{d\theta} = -4\cos\theta\sin 2\theta - 2\sin\theta\cos 2\theta$$
$$r = 2\cos 2\theta \Rightarrow y = 2\sin\theta\cos 2\theta, \quad \frac{dy}{d\theta} = -4\sin\theta\sin 2\theta + 2\cos\theta\cos 2\theta$$
$$\frac{dx}{d\theta} = -2\sin\theta(6\cos^2\theta - 1) = 0 \Rightarrow \sin\theta = 0 \text{ or } \cos\theta = \pm\frac{1}{\sqrt{6}}$$

When $\theta = 0$, $x = 2$; when $\theta = \pi$, $x = -2$. At the other values, the corresponding x-coordinates are $-\frac{2\sqrt{6}}{9}$ and $\frac{2\sqrt{6}}{9}$. So the tangent lines are $x = -2$, $x = 2$, $x = -\frac{2\sqrt{6}}{9}$ and $x = \frac{2\sqrt{6}}{9}$.

5 d)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = a - a\sin\theta \Rightarrow x = a\cos\theta - \frac{a}{2}\sin 2\theta, \quad \frac{dx}{d\theta} = -a\sin\theta + a\cos 2\theta$$
$$r = a - a\sin\theta \Rightarrow y = a\sin\theta - a\sin^2\theta, \quad \frac{dy}{d\theta} = a\cos\theta + 2a\sin\theta\cos\theta$$
$$-a\sin\theta + a\cos 2\theta = -a\sin\theta + 2a(1 - 2\sin^2\theta)$$
$$= -a\sin\theta + 2a - 2a\sin^2\theta$$
$$-a(2\sin\theta + 1)(\sin\theta - 1) = 0$$
$$\sin\theta = -\frac{1}{2} \quad \text{or} \quad \sin\theta = 1$$
$$\theta = \frac{\pi}{6}, \quad \theta = \frac{5\pi}{6}, \quad \text{or} \quad \theta = \frac{\pi}{2}$$

When $\theta = -\frac{\pi}{6}$, $x = \frac{3\sqrt{3}}{4}a$. When $\theta = -\frac{5\pi}{6}$, $x = -\frac{3\sqrt{3}}{4}a$. When $\theta = \frac{\pi}{2}$, $x = 0$ but $\dfrac{dy}{d\theta} = 0$, so this is a cusp point. So the tangent lines are $x = \frac{3\sqrt{3}}{4}a$ and $x = -\frac{3\sqrt{3}}{4}a$.

5 e)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = a + a\cos\theta \Rightarrow x = a\cos\theta + a\cos^2\theta, \quad \frac{dx}{d\theta} = -a\sin\theta - 2a\cos\theta\sin\theta$$
$$r = a + a\cos\theta \Rightarrow y = a\sin\theta + \frac{a}{2}\sin 2\theta, \quad \frac{dy}{d\theta} = a\cos\theta + a\cos 2\theta$$
$$-a\sin\theta(2\cos\theta + 1) = 0 \Rightarrow \theta = 0, \pi, -\frac{2\pi}{3}, \frac{2\pi}{3}$$

At $\theta = \pi$, $x = 0$ but $\dfrac{dy}{d\theta} = 0$, so this is a cusp. At $\theta = 0$, $x = 2a$; at $\theta = \pm\frac{2\pi}{3}$, $x = -\frac{1}{4}a$. So the tangent lines are $x = 2a$ and $x = -\frac{1}{4}a$.

5 f)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = a\sin 2\theta \Rightarrow x = 2a\cos^2\theta\sin\theta, \quad \frac{dx}{d\theta} = 2a\cos^3\theta - 4a\cos\theta\sin^2\theta$$
$$r = a\sin 2\theta \Rightarrow y = 2a\sin^2\theta\cos\theta, \quad \frac{dy}{d\theta} = -2a\sin^3\theta + 4a\sin\theta\cos^2\theta$$
$$-2a\cos\theta(1 - 3\sin^2\theta) = 0 \Rightarrow \cos\theta = 0 \text{ or } \sin\theta = \pm\frac{1}{\sqrt{3}}$$

At $\theta = -\frac{\pi}{2}$, $x = 0$ but $\dfrac{dy}{d\theta} = 0$, so this is a cusp. At the other values the x-coordinates are $\frac{4\sqrt{3}}{9}a$ and $-\frac{4\sqrt{3}}{9}a$. So the tangent lines are $x = \frac{4\sqrt{3}}{9}a$ and $x = -\frac{4\sqrt{3}}{9}a$.

5 g)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = e^\theta \Rightarrow x = e^\theta\cos\theta, \quad \frac{dx}{d\theta} = e^\theta\cos\theta - e^\theta\sin\theta$$
$$r = e^\theta \Rightarrow y = e^\theta\sin\theta, \quad \frac{dy}{d\theta} = e^\theta\sin\theta + e^\theta\cos\theta$$
$$e^\theta(\cos\theta - \sin\theta) = 0 \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4} \text{ or } \theta = -\frac{3\pi}{4}$$
$$x\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}e^{\pi/4}, \qquad x\left(-\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}e^{-3\pi/4}$$

So the tangent lines are $x = \frac{\sqrt{2}}{2}e^{\pi/4}$ and $x = -\frac{\sqrt{2}}{2}e^{-3\pi/4}$.

5 h)

A tangent perpendicular to the initial line $\theta = 0$ requires $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = 1 + \tan\theta \Rightarrow x = \cos\theta + \cos\theta\tan\theta, \quad \frac{dx}{d\theta} = -\sin\theta + \cos\theta\sec^2\theta - \sin\theta\tan\theta$$
$$r = 1 + \tan\theta \Rightarrow y = \sin\theta + \sin\theta\tan\theta, \quad \frac{dy}{d\theta} = \cos\theta + \sin\theta\sec^2\theta + \cos\theta\tan\theta$$
$$\frac{dx}{d\theta} = 0 \Rightarrow \cos\theta(\cos\theta - \sin\theta) = 0 \Rightarrow \tan\theta = 1 \quad (\cos\theta \neq 0)$$
$$\therefore \theta = \frac{\pi}{4} \text{ or } \theta = -\frac{3\pi}{4}$$
$$x\left(\frac{\pi}{4}\right) = \sqrt{2}, \qquad x\left(-\frac{3\pi}{4}\right) = -\sqrt{2}$$

So the tangent lines are $x = \sqrt{2}$ and $x = -\sqrt{2}$.

6 a)

Tangent parallel to the initial line $\theta = 0$: $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = 2 + \sqrt{3}\cos\theta \Rightarrow y = r\sin\theta = 2\sin\theta + \sqrt{3}\sin\theta\cos\theta$$
$$\frac{dy}{d\theta} = 2\cos\theta - \sqrt{3}\sin^2\theta + \sqrt{3}\cos 2\theta$$

Considering $\dfrac{dy}{d\theta} = 0$ at $\theta = \phi$ gives

$$2\cos\phi - \sqrt{3}(1 - \cos 2\phi) + \sqrt{3}\cos 2\phi = 0$$
$$2\sqrt{3}\cos^2\phi + 2\cos\phi - \sqrt{3} = 0$$
$$\cos\phi = \frac{-2 \pm \sqrt{4+24}}{4\sqrt{3}} = \frac{-1 \pm \sqrt{7}}{2\sqrt{3}}$$

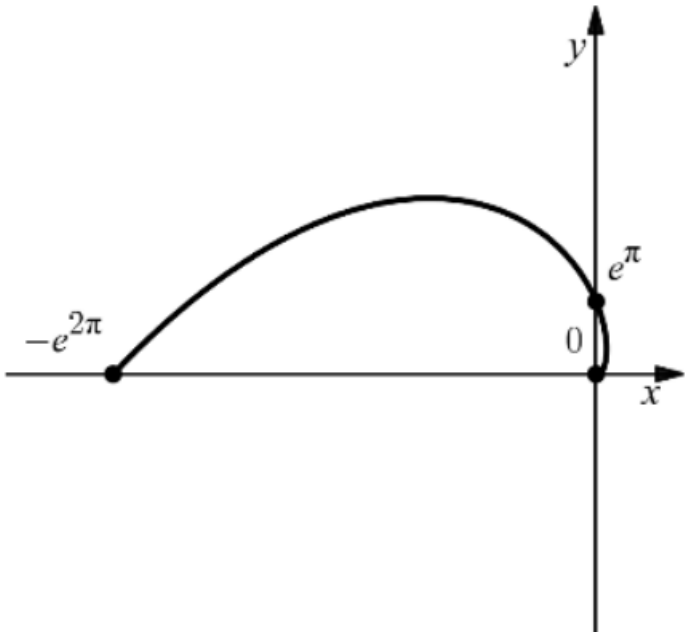
Since $0 < \phi < \frac{\pi}{2}$, $\cos\phi > 0$. Hence $\cos\phi = \dfrac{-1+\sqrt{7}}{2\sqrt{3}}$.

6 b)

$$r = 2 + \sqrt{3}\left(\frac{-1+\sqrt{7}}{2\sqrt{3}}\right) = 2 + \frac{-1+\sqrt{7}}{2} = \frac{3+\sqrt{7}}{2}.$$

7 a)

Sketch of $r = e^{2\theta}$ for $0 \le \theta \le \pi$.



7 b)

$$e^{2\pi}$$

7 c)

Tangent perpendicular to the initial line $\theta = 0$: $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = e^{2\theta} \Rightarrow x = e^{2\theta}\cos\theta, \quad \frac{dx}{d\theta} = -e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta$$
$$e^{2\theta}(2\cos\theta - \sin\theta) = 0 \Rightarrow \tan\theta = 2 \Rightarrow \theta = \arctan(2)$$

So $r = e^{2\arctan(2)}$ and the distance is $e^{2\arctan(2)} = 9.15$ units (3 SF).

7 d)

Tangent parallel to the initial line $\theta = 0$: $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = e^{2\theta} \Rightarrow y = e^{2\theta}\sin\theta, \quad \frac{dy}{d\theta} = e^{2\theta}\cos\theta + 2e^{2\theta}\sin\theta$$
$$e^{2\theta}(\cos\theta + 2\sin\theta) = 0 \Rightarrow \tan\theta = -\frac{1}{2}$$
$$0 < \theta < \pi \Rightarrow \theta = \pi - \arctan\left(\frac{1}{2}\right)$$

So $r = e^{2(\pi - \arctan(1/2))}$ and the distance is $e^{2(\pi-\arctan(1/2))} = 212$ units (3 SF).

8)

Find the position of line AD. Tangent perpendicular to the initial line $\theta = 0$: $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = \frac{1}{2}\sin 2\theta \Rightarrow x = \frac{1}{2}\sin 2\theta\cos\theta = \sin\theta\cos^2\theta$$
$$\frac{dx}{d\theta} = -2\sin^2\theta\cos\theta + \cos^3\theta$$
$$\frac{dx}{d\theta} = \cos\theta(\cos^2\theta - 2\sin^2\theta) = 0$$

Thus $\cos\theta = 0$ or $\tan\theta = \pm\dfrac{1}{\sqrt{2}}$. Considering $\tan\theta = \dfrac{1}{\sqrt{2}}$ gives $\sin\theta = \dfrac{1}{\sqrt{3}}$ and $\cos\theta = \sqrt{\dfrac{2}{3}}$, so $x = \dfrac{1}{\sqrt{3}} \times \dfrac{2}{3} = \dfrac{2\sqrt{3}}{9}$. Therefore the area of the square is $\left(2 \times \dfrac{2\sqrt{3}}{9}\right)^2 = \dfrac{16}{27}$ units^2.

9 a)

Tangent parallel to the initial line $\theta = 0$: $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = a + a\cos\theta \Rightarrow y = a\sin\theta + \frac{a}{2}\sin 2\theta, \quad \frac{dy}{d\theta} = a\cos\theta + a\cos 2\theta$$
$$a\cos\theta + a\cos 2\theta = a(\cos\theta + 1)(2\cos\theta - 1) = 0$$
$$\cos\theta = -1 \quad \text{or} \quad \cos\theta = \frac{1}{2}$$

Hence $\theta = \pi, \frac{\pi}{3}, \frac{5\pi}{3}$. When $\theta = \pi$, $r = 0$. When $\theta = \frac{\pi}{3}$, $r = \frac{3}{2}a$, and when $\theta = \frac{5\pi}{3}$, $r = \frac{3}{2}a$. So $P\left(\frac{3}{2}a, \frac{\pi}{3}\right)$ and $Q\left(\frac{3}{2}a, \frac{5\pi}{3}\right)$.

9 b)

By symmetry, consider $\theta = \frac{\pi}{3}$, where $y = \frac{3\sqrt{3}}{4}a$. The distance is $2 \times \frac{3\sqrt{3}}{4}a = 12$, so $a = \dfrac{8}{\sqrt{3}} = \dfrac{8\sqrt{3}}{3}$.

10 a)

Tangent perpendicular to the initial line $\theta = 0$: $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = a - a\sin\theta \Rightarrow x = a\cos\theta - \frac{a}{2}\sin 2\theta, \quad \frac{dx}{d\theta} = -a\sin\theta - a\cos 2\theta$$
$$-a\sin\theta - a\cos 2\theta = a(\sin\theta - 1)(2\sin\theta + 1) = 0$$

$$\sin\theta = 1 \quad \text{or} \quad \sin\theta = -\frac{1}{2}$$

Thus $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$. When $\theta = \frac{\pi}{2}$, $r = 0$. At the other values, $r = \frac{3}{2}a$. So $P\left(\frac{3}{2}a, \frac{7\pi}{6}\right)$ and $Q\left(\frac{3}{2}a, \frac{11\pi}{6}\right)$.

10 b)

By symmetry, consider $\theta = \frac{7\pi}{6}$, where $x = -\frac{3\sqrt{3}}{4}a$. The distance is $2 \times \frac{3\sqrt{3}}{4}a = 5$, so $a = \dfrac{10}{3\sqrt{3}} = \dfrac{10\sqrt{3}}{9}$.

11 a)

Tangent perpendicular to the initial line $\theta = 0$: $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

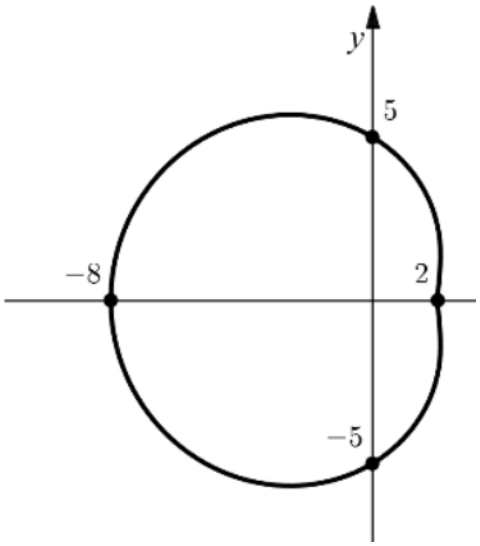
$$r = k - 3\cos\theta \Rightarrow x = k\cos\theta - 3\cos^2\theta, \quad \frac{dx}{d\theta} = -k\sin\theta + 6\cos\theta\sin\theta$$
$$-k\sin\theta + 6\cos\theta\sin\theta = \sin\theta(6\cos\theta - k) = 0$$

So $\sin\theta = 0$ or $\cos\theta = \dfrac{k}{6}$. For $\sin\theta = 0$ we obtain two points, at $\theta = 0$ and $\theta = \pi$. The other two points come from solving $\cos\theta = \dfrac{k}{6}$, so $-1 < \dfrac{k}{6} < 1$. That is, $-6 < k < 6$. But since $k > 3$, $3 < k < 6$ as required.

11 b)

Sketch of $r = 5 - 3\cos\theta$ for $0 \leq \theta < 2\pi$.

.



12 a)

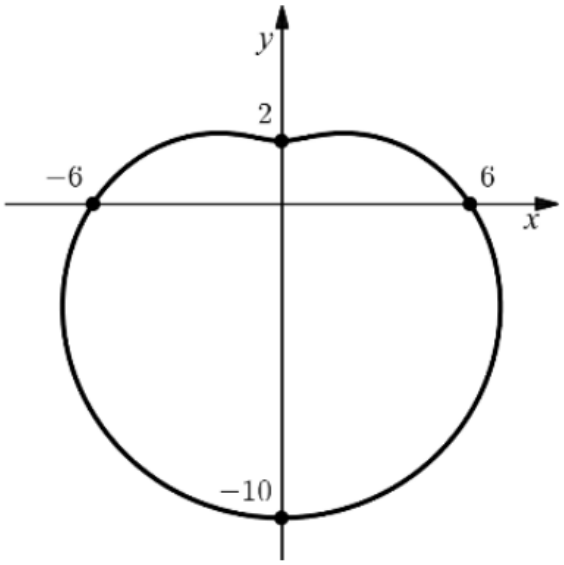
Tangent parallel to the initial line $\theta = 0$: $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = k - 4\sin\theta \Rightarrow y = k\sin\theta - 4\sin^2\theta, \quad \frac{dy}{d\theta} = k\cos\theta - 8\cos\theta\sin\theta$$
$$k\cos\theta - 8\cos\theta\sin\theta = \cos\theta(k - 8\sin\theta) = 0$$

So $\cos\theta = 0$ or $\sin\theta = \dfrac{k}{8}$. For $\cos\theta = 0$ we obtain two points, at $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$. The other two points come from solving $\sin\theta = \dfrac{k}{8}$, so $-1 < \dfrac{k}{8} < 1$. That is, $-8 < k < 8$. But since $k > 4$, $4 < k < 8$ as required.

12 b)

Sketch of $r = 6 - 4\sin\theta$ for $0 \leq \theta < 2\pi$.



13 a)

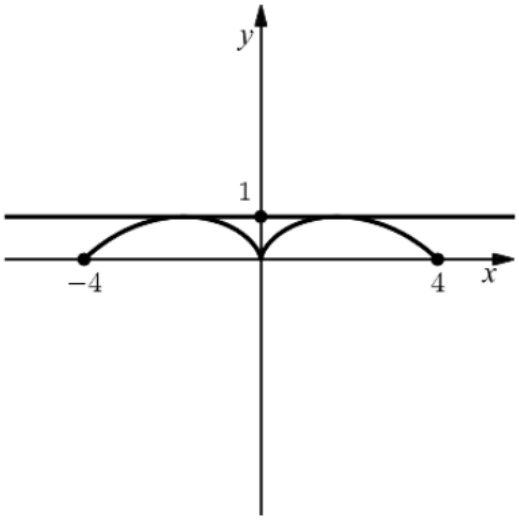
Tangent parallel to the initial line $\theta = 0$: $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

$$r = 4 - 4\sin\theta \Rightarrow y = 4\sin\theta - 4\sin^2\theta, \quad \frac{dy}{d\theta} = 4\cos\theta - 8\sin\theta\cos\theta$$
$$r = 4 - 4\sin\theta \Rightarrow x = 4\cos\theta - 2\sin 2\theta, \quad \frac{dx}{d\theta} = -4\sin\theta - 4\cos 2\theta$$
$$4\cos\theta(1 - 2\sin\theta) = 0 \Rightarrow \cos\theta = 0 \text{ or } \sin\theta = \frac{1}{2}$$

Thus $\theta = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$. When $\theta = \frac{\pi}{2}$, $y = 0$ and $\dfrac{dx}{d\theta} = 0$, so there is a cusp. When $\theta = \frac{\pi}{6}$ and $\frac{5\pi}{6}$, $y = 1$. So the tangent line L is $y = 1$.

13 b)

Sketch the curve C and the line L on the same axes.



13 c)

The line L and the curve C intersect when $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$. When $\theta = \frac{\pi}{6}$, $x = 4\cos\frac{\pi}{6} - 2\sin\frac{\pi}{3} = \sqrt{3}$. So the area of the rectangle under L is $1 \times 2\sqrt{3} = 2\sqrt{3}$. The area under the polar curve is

$$\begin{aligned}
\frac{1}{2}\int_{\pi/6}^{5\pi/6} (4 - 4\sin\theta)^2\, d\theta &= 8\int_{\pi/6}^{5\pi/6} (1 - 2\sin\theta + \sin^2\theta)\, d\theta \\
&= 8\int_{\pi/6}^{5\pi/6} 1\, d\theta - 16\int_{\pi/6}^{5\pi/6} \sin\theta\, d\theta + 4\int_{\pi/6}^{5\pi/6} (1 - \cos 2\theta)\, d\theta \\
&= 8\,[\theta]_{\pi/6}^{5\pi/6} + 16\,[\cos\theta]_{\pi/6}^{5\pi/6} + 4\left[\theta - \frac{1}{2}\sin 2\theta\right]_{\pi/6}^{5\pi/6} \\
&= \frac{16\pi}{3} - 16\sqrt{3} + 2\sqrt{3} + \frac{8\pi}{3} = 8\pi - 14\sqrt{3}.
\end{aligned}$$

The area of the triangle between $(\sqrt{3}, 1)$, $(-\sqrt{3}, 1)$ and $(0, 0)$ is $\frac{1}{2}(2\sqrt{3})(1) = \sqrt{3}$. So the area required is $\sqrt{3} - (8\pi - 14\sqrt{3}) = (15\sqrt{3} - 8\pi)$ units^2.

14 a)

Tangent perpendicular to the initial line $\theta = 0$: $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$r = 2 + \tan\theta \Rightarrow x = 2\cos\theta + \cos\theta\tan\theta, \quad \frac{dx}{d\theta} = -2\sin\theta + \cos\theta\sec^2\theta - \sin\theta\tan\theta$$

$$-2\sin\theta + \frac{1}{\cos\theta} - \frac{\sin^2\theta}{\cos\theta} = 0$$

$$-2\sin\theta\cos\theta + 1 - \sin^2\theta = \cos\theta(\cos\theta - 2\sin\theta) = 0$$

So $\cos\theta = 0$ or $\tan\theta = \frac{1}{2}$ (with $\cos\theta \neq 0$). Thus $\tan\theta = \frac{1}{2}$ and the angle is acute, so $\sin\theta = \dfrac{1}{\sqrt{5}}$ and $\cos\theta = \dfrac{2}{\sqrt{5}}$.

14 b)

$$x = 2\cos\theta + \cos\theta\tan\theta = 2\left(\frac{2}{\sqrt{5}}\right) + \frac{2}{\sqrt{5}} \times \frac{1}{2} = \sqrt{5}.$$

14 c)

At $\phi = \arctan(1/2)$, $r = 2 + \tan\phi = \frac{5}{2}$. The area of the triangle is $\frac{1}{2} \times \frac{1}{2} \times \sqrt{5} \times \frac{5}{2\sqrt{5}} = \frac{5}{8}$.

$$\begin{aligned}\frac{1}{2}\int (2+\tan\theta)^2\,d\theta &= 2\int 1\,d\theta + 2\int \tan\theta\,d\theta + \frac{1}{2}\int (\sec^2\theta - 1)\,d\theta \\ &= \frac{3}{2}\theta + 2\ln|\sec\theta| + \frac{1}{2}\tan\theta + C.\end{aligned}$$

Therefore, where $\phi = \arctan(1/2)$,

$$\begin{aligned}\frac{1}{2}\int_0^{\phi} (2+\tan\theta)^2\,d\theta &= \left(\frac{3}{2}\phi + 2\ln|\sec\phi| + \frac{1}{2}\tan\phi\right) - \left(0 + 2\ln|\sec 0| + \frac{1}{2}\tan 0\right) \\ &= \frac{3}{2}\arctan\left(\frac{1}{2}\right) + 2\ln\left|\frac{\sqrt{5}}{2}\right| + \frac{1}{4}.\end{aligned}$$

The required area is $\frac{5}{8} - \left(\frac{3}{2}\arctan\left(\frac{1}{2}\right) + 2\ln\left|\frac{\sqrt{5}}{2}\right| + \frac{1}{4}\right) = 0.0814$ units^2 (3 SF).

15 a)

Tangent perpendicular to the initial line $\theta = 0$: $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

$$\begin{aligned}r = 6\sqrt{3} - 2\cos 3\theta \Rightarrow x &= 6\sqrt{3}\cos\theta - 2\cos\theta\cos 3\theta \\ \frac{dx}{d\theta} &= -6\sqrt{3}\sin\theta + 2\sin\theta\cos 3\theta + 6\cos\theta\sin 3\theta\end{aligned}$$

When $\theta = \frac{\pi}{6}$,

$$\frac{dx}{d\theta} = -6\sqrt{3}\sin\frac{\pi}{6} + 2\sin\frac{\pi}{6}\cos\frac{\pi}{2} + 6\cos\frac{\pi}{6}\sin\frac{\pi}{2} = -3\sqrt{3} + 0 + 3\sqrt{3} = 0,$$

as required.

15 b)

Find the area above the axes and double it. When $\theta = \frac{\pi}{6}$, $r = 6\sqrt{3}$ and $x = 6\sqrt{3}\cos\frac{\pi}{6} - 2\cos\frac{\pi}{6}\cos\frac{\pi}{2} = 9$. The area of the triangle is $\frac{1}{2}(6\sqrt{3})(9)\sin\frac{\pi}{6} = \frac{27\sqrt{3}}{2}$.

$$\begin{aligned}\frac{1}{2}\int_0^{\pi/6} (6\sqrt{3} - 2\cos 3\theta)^2\,d\theta &= \frac{1}{2}\int_0^{\pi/6} (108 - 24\sqrt{3}\cos 3\theta + 4\cos^2 3\theta)\,d\theta \\ &= 54\int_0^{\pi/6} 1\,d\theta - 12\sqrt{3}\int_0^{\pi/6}\cos 3\theta\,d\theta + \int_0^{\pi/6}(1+\cos 6\theta)\,d\theta \\ &= 54\theta - 4\sqrt{3}\sin 3\theta + \theta + \frac{1}{6}\sin 6\theta\Big|_0^{\pi/6} = \frac{55\pi}{6} - 4\sqrt{3}.\end{aligned}$$

Thus the triangle minus the curved area is $\frac{27\sqrt{3}}{2} - \left(\frac{55\pi}{6} - 4\sqrt{3}\right) = \frac{35\sqrt{3}}{2} - \frac{55\pi}{6}$. Doubling this area gives $\left(35\sqrt{3} - \frac{55\pi}{3}\right)$ units^2.

16 a)

$$r = 3\cos\theta, \quad \frac{dr}{d\theta} = -3\sin\theta.$$

Maximum r is when $\dfrac{dr}{d\theta} = 0$, so $-3\sin\theta = 0$. For $0 \le \theta \le \frac{\pi}{2}$ this is when $\theta = 0$, and $r = 3\cos 0 = 3$. At the interval endpoints $r = 3$ and $r = 0$. Maximum is $r = 3$ when $\theta = 0$.

16 b)

$$r = 2 + \sin\theta, \quad \frac{dr}{d\theta} = \cos\theta.$$

Maximum r is when $\dfrac{dr}{d\theta} = 0$, so $\cos\theta = 0$. For $0 \le \theta \le \pi$ this is when $\theta = \frac{\pi}{2}$, and $r = 2 + \sin\frac{\pi}{2} = 3$. At the interval endpoints $r = 2$ and $r = 2$. Maximum is $r = 3$ when $\theta = \frac{\pi}{2}$.

16 c)

$$r = 5\cos 2\theta, \quad \frac{dr}{d\theta} = -10\sin 2\theta.$$

Maximum r is when $\dfrac{dr}{d\theta} = 0$, so $-10\sin 2\theta = 0$. For $0 \le \theta \le \frac{\pi}{4}$ this is when $\theta = 0$, and $r = 5\cos 0 = 5$. At the interval endpoints $r = 5$ and $r = 0$. Maximum is $r = 5$ when $\theta = 0$.

16 d)

$$r = \ln(1 + \sin\theta), \quad \frac{dr}{d\theta} = \frac{\cos\theta}{1 + \sin\theta}.$$

Maximum r is when $\dfrac{dr}{d\theta} = 0$, so $\cos\theta = 0$. For $0 \le \theta < \pi$ this is when $\theta = \frac{\pi}{2}$, and $r = \ln(1+\sin\frac{\pi}{2}) = \ln 2$. At the interval endpoints $r = \ln 1 = 0$ and $r = \ln 1 = 0$. Maximum is $r = \ln 2$ when $\theta = \frac{\pi}{2}$.

17 a)

$$r = 4\cos(2\theta)\cosh\left(\frac{\theta}{2}\right).$$

When $\theta = \frac{\pi}{4}$, $r = 4\cos(\frac{\pi}{2})\cosh(\frac{\pi}{8}) = 0$, and when $\theta = \frac{3\pi}{4}$, $r = 4\cos(\frac{3\pi}{2})\cosh(\frac{3\pi}{8}) = 0$. Since $\cos 2\theta = 0$ only at $\frac{\pi}{4}$ and $\frac{3\pi}{4}$, the curve starts and ends at the pole. On $\frac{\pi}{4} < \theta < \frac{3\pi}{4}$, $\cos 2\theta < 0$ and $\cosh(\frac{\theta}{2}) > 0$, so $r < 0$ throughout the interval, and the curve consists of only one loop.

17 b)

$$r = 4\cos(2\theta)\cosh\left(\frac{\theta}{2}\right), \quad \frac{dr}{d\theta} = 2\cos(2\theta)\sinh\left(\frac{\theta}{2}\right) - 8\sin(2\theta)\cosh\left(\frac{\theta}{2}\right).$$

The interval endpoints give $r = 0$. The maximum r occurs when $\dfrac{dr}{d\theta} = 0$ at $\theta = \phi$:

$$2\cos(2\phi)\sinh\left(\frac{\phi}{2}\right) - 8\sin(2\phi)\cosh\left(\frac{\phi}{2}\right) = 0$$

$$\cos(2\phi)\sinh\left(\frac{\phi}{2}\right) = 4\sin(2\phi)\cosh\left(\frac{\phi}{2}\right)$$

$$\begin{aligned}
\frac{\sinh(\phi/2)}{\cosh(\phi/2)} &= 4\tan 2\phi \\
e^{\phi/2} - e^{-\phi/2} &= 4\tan 2\phi(e^{\phi/2} + e^{-\phi/2}) \\
e^{\phi/2}(1 - 4\tan 2\phi) &= e^{-\phi/2}(1 + 4\tan 2\phi) \\
e^{\phi} &= \frac{1 + 4\tan 2\phi}{1 - 4\tan 2\phi} \\
\phi &= \ln\left(\frac{1 + 4\tan 2\phi}{1 - 4\tan 2\phi}\right),
\end{aligned}$$

as required.

17 c)

When $\phi = 1.65488$, $r = 4\cos(2\times 1.65488)\cosh(1.65488/2) = -5.372$ (4 SF). So the distance is 5.372 units.

18)

$$\frac{dx}{d\theta} = -r\sin\theta + \frac{dr}{d\theta}\cos\theta, \quad \frac{dy}{d\theta} = r\cos\theta + \frac{dr}{d\theta}\sin\theta$$
$$\frac{dy}{dx} = \frac{\dfrac{dy}{d\theta}}{\dfrac{dx}{d\theta}} = \frac{r\cos\theta + \dfrac{dr}{d\theta}\sin\theta}{-r\sin\theta + \dfrac{dr}{d\theta}\cos\theta} = \frac{\dfrac{dr}{d\theta}\sin\theta + r\cos\theta}{\dfrac{dr}{d\theta}\cos\theta - r\sin\theta},$$

as required.

19 a)

$$r = 2\theta, \quad \frac{dr}{d\theta} = 2, \quad \frac{dy}{dx} = \frac{\dfrac{dr}{d\theta}\sin\theta + r\cos\theta}{\dfrac{dr}{d\theta}\cos\theta - r\sin\theta} = \frac{2\sin\theta + 2\theta\cos\theta}{2\cos\theta - 2\theta\sin\theta}.$$

Therefore

$$\left.\frac{dy}{dx}\right|_{\theta=\pi/2} = \frac{2\sin(\pi/2) + \pi\cos(\pi/2)}{2\cos(\pi/2) - \pi\sin(\pi/2)} = -\frac{2}{\pi}.$$

19 b)

$$r = 2\cos\theta, \quad \frac{dr}{d\theta} = -2\sin\theta, \quad \frac{dy}{dx} = \frac{-2\sin^2\theta + 2\cos^2\theta}{-2\sin\theta\cos\theta - 2\cos\theta\sin\theta}.$$

Therefore

$$\left.\frac{dy}{dx}\right|_{\theta=\pi/6} = \frac{-2(1/2)^2 + 2(\sqrt{3}/2)^2}{-2(1/2)(\sqrt{3}/2) - 2(\sqrt{3}/2)(1/2)} = \frac{-2+2}{\sqrt{3}+\sqrt{3}} = -\frac{\sqrt{3}}{3}.$$

19 c)

$$r = 1 + 2\sin\theta, \quad \frac{dr}{d\theta} = 2\cos\theta, \quad \frac{dy}{dx} = \frac{2\cos\theta\sin\theta + (1+2\sin\theta)\cos\theta}{2\cos^2\theta - (1+2\sin\theta)\sin\theta}.$$

Thus

$$\left.\frac{dy}{dx}\right|_{\theta=3\pi/4} = \frac{-1-1-2(\sqrt{2}/2)}{1-1-2(\sqrt{2}/2)} = 1 + 2\sqrt{2}.$$

20 a)

$$r = \theta \sin\theta, \quad \frac{dr}{d\theta} = \theta\cos\theta + \sin\theta.$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{(\theta\cos\theta + \sin\theta)\sin\theta + \theta\sin\theta\cos\theta}{(\theta\cos\theta + \sin\theta)\cos\theta - \theta\sin^2\theta} \\ &= \frac{2\theta\cos\theta\sin\theta + \sin^2\theta}{\theta(\cos^2\theta - \sin^2\theta) + \sin\theta\cos\theta} \\ &= \frac{2\theta\cos\theta\sin\theta + \sin^2\theta}{\theta\cos 2\theta + \sin\theta\cos\theta}.\end{aligned}$$

Multiplying numerator and denominator by 2 and using $2\cos\theta\sin\theta = \sin 2\theta$ gives

$$\frac{dy}{dx} = \frac{2\theta\sin 2\theta + 2\sin^2\theta}{2\theta\cos 2\theta + \sin 2\theta},$$

as required.

20 b)

Solving the numerator set to zero:

$$\begin{gathered}4\theta\cos\theta\sin\theta + 2\sin^2\theta = 0 \\ \sin\theta(2\theta\cos\theta + \sin\theta) = 0 \\ \sin\theta = 0 \quad \text{or} \quad \tan\theta = -2\theta\end{gathered}$$

For $0 \le \theta < \pi$, $\sin\theta = 0$ has one solution, $\theta = 0$. The equation $\tan\theta = -2\theta$ has two solutions, one of which is $\theta = 0$. Solving the denominator set to zero:

$$2\theta\cos 2\theta + \sin 2\theta = 0 \Rightarrow \tan 2\theta = -2\theta.$$

For $0 \le \theta < \pi$, $\tan 2\theta = -2\theta$ has three solutions, one of which is $\theta = 0$. So there is one horizontal tangent and two vertical tangents. When $\theta = 0$, $r = 0$, and when $\theta = \pi$, $r = 0$; since there is one horizontal tangent, there is one loop.

21 a)

$$r = e^{k\theta}, \quad \frac{dr}{d\theta} = ke^{k\theta}.$$

$$\frac{dy}{dx} = \frac{ke^{k\theta}\sin\theta + e^{k\theta}\cos\theta}{ke^{k\theta}\cos\theta - e^{k\theta}\sin\theta} = \frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta},$$

since $e^{k\theta} \neq 0$.

21 b)

The line OP makes an angle of θ with the x-axis. Let A be the point at which the tangent crosses the positive x-axis, so $\phi = \angle OPA$. Further let α be the obtuse angle that the tangent makes with the x-axis at A. By geometry, $\theta + \phi = \alpha$, so $\phi = \alpha - \theta$ and

$$\tan\phi = \tan(\alpha - \theta) = \frac{\tan\alpha - \tan\theta}{1 + \tan\alpha\tan\theta}.$$

Since $\tan\alpha = \dfrac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta}$,

$$\tan\alpha - \tan\theta = \frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta} - \frac{\sin\theta}{\cos\theta}$$

$$
\begin{aligned}
&= \frac{\cos\theta(k\sin\theta+\cos\theta)-\sin\theta(k\cos\theta-\sin\theta)}{\cos\theta(k\cos\theta-\sin\theta)}\\
&= \frac{k\sin\theta\cos\theta+\cos^2\theta-k\cos\theta\sin\theta+\sin^2\theta}{\cos\theta(k\cos\theta-\sin\theta)}\\
&= \frac{1}{\cos\theta(k\cos\theta-\sin\theta)},\\
1+\tan\alpha\tan\theta &= 1+\frac{(k\sin\theta+\cos\theta)\sin\theta}{(k\cos\theta-\sin\theta)\cos\theta}\\
&= \frac{k\cos^2\theta-\cos\theta\sin\theta+k\sin^2\theta+\sin\theta\cos\theta}{\cos\theta(k\cos\theta-\sin\theta)}\\
&= \frac{k}{\cos\theta(k\cos\theta-\sin\theta)}.
\end{aligned}
$$

Therefore $\tan\phi = \dfrac{1}{k}$, as required.

22)

$$r = \sin 2\theta, \quad \frac{dr}{d\theta} = 2\cos 2\theta.$$

Consider the line $\theta = \frac{\pi}{4}$. Let $\theta_1 = \frac{\pi}{4}+\phi$ and $\theta_2 = \frac{\pi}{4}-\phi$. At point 1,

$$r_1 = \sin\left(\frac{\pi}{2}+2\phi\right) = \cos 2\phi, \quad \frac{dr_1}{d\theta_1} = 2\cos\left(\frac{\pi}{2}+2\phi\right) = -2\sin 2\phi$$
$$\sin\theta_1 = \frac{\sqrt{2}}{2}(\cos\phi+\sin\phi), \quad \cos\theta_1 = \frac{\sqrt{2}}{2}(\cos\phi-\sin\phi)$$

$$
\begin{aligned}
\left.\frac{dy}{dx}\right|_{\theta_1} &= \frac{-2\sin 2\phi\sin\theta_1+\cos 2\phi\cos\theta_1}{-2\sin 2\phi\cos\theta_1-\cos 2\phi\sin\theta_1}\\
&= \frac{\begin{matrix}-2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi)\\ +\sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)\end{matrix}}{\begin{matrix}-2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi)\\ -\sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)\end{matrix}}\\
&= \frac{\begin{matrix}2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi)\\ -\sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)\end{matrix}}{\begin{matrix}2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi)\\ +\sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)\end{matrix}}.
\end{aligned}
$$

At point 2,

$$r_2 = \sin\left(\frac{\pi}{2}-2\phi\right) = \cos 2\phi, \quad \frac{dr_2}{d\theta_2} = 2\cos\left(\frac{\pi}{2}-2\phi\right) = 2\sin 2\phi$$
$$\sin\theta_2 = \frac{\sqrt{2}}{2}(\cos\phi-\sin\phi), \quad \cos\theta_2 = \frac{\sqrt{2}}{2}(\cos\phi+\sin\phi)$$

$$
\begin{aligned}
\left.\frac{dy}{dx}\right|_{\theta_2} &= \frac{2\sin 2\phi\sin\theta_2+\cos 2\phi\cos\theta_2}{2\sin 2\phi\cos\theta_2-\cos 2\phi\sin\theta_2}\\
&= \frac{\begin{matrix}2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi)\\ +\sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)\end{matrix}}{\begin{matrix}2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi)\\ -\sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)\end{matrix}}.
\end{aligned}
$$

Comparing the expressions shows that the gradients at points equidistant from $\theta = \frac{\pi}{4}$ are reciprocals, so they are symmetrically related as reflections in the line $\theta = \frac{\pi}{4}$.